

UbiPower18000_16x15.36TB_NVMe_4xNDR400 _3xStorageNode_1hosts256GBMem

System Description

System Name: UbiPower18000_16x15.36TB_NVMe_4xNDR400_3xStorageNode_1hosts256GBMem

Submitter Name: UBIX TECHNOLOGY CO., LTD

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Benchmark Version: MLPerf Storage v3.0

Solution Type: Proprietary - UBIXFS(distributed filesystem)

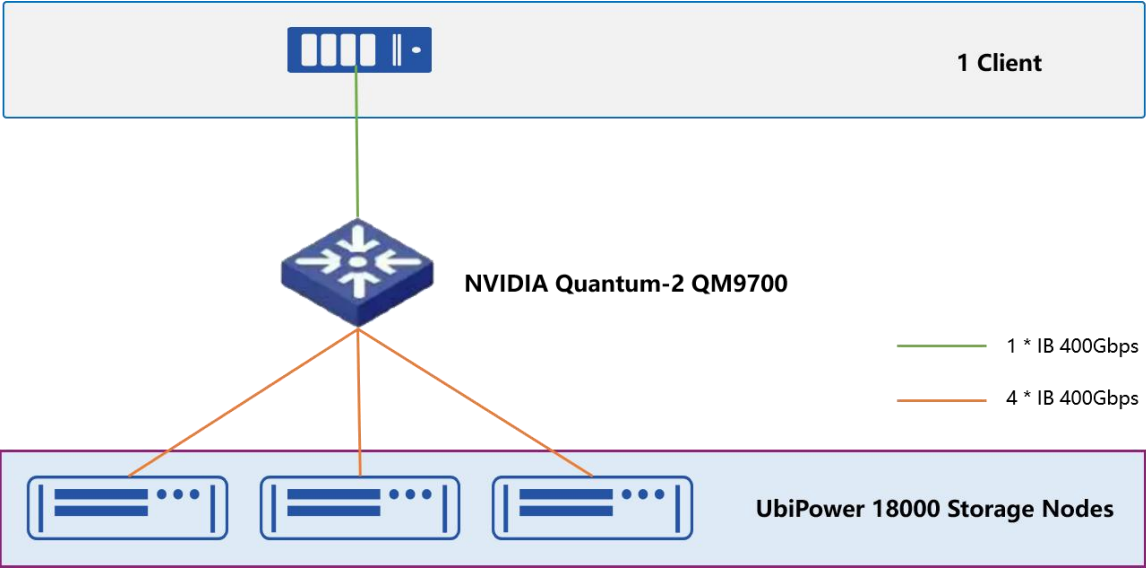
Submission Division: CLOSED

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1. Hardware

The hardware is composed of 1 client and 3 servers connected to the same switch.



1.1. Client

The client environment consists of commercial x86 server nodes, each equipped with dual CPU sockets and 256 GB of system memory. Storage I/O traffic between the clients and the storage server is carried over InfiniBand using the links dedicated to storage.

The detailed configuration of the client is shown in the following table:

Client Node components	Configuration Description	Client Node Count
Client CPU	Intel(R) Xeon(R) 6966P-C (*2)	1
Client Memory	256GB DDR5 (64GB *4)	
Client Network Type	Mellanox CX7 Single-port IB NDR 400Gb (*1)	

1.2. Server

The server is a commercial UbiPower 18000 distributed all-flash storage system based on x86 architecture, designed for high performance, low latency, and high scalability.

The detailed configuration of the servers is shown in the following table:

Server components	Node	Configuration Description	Server Node Count
Server CPU		INTEL(R) XEON(R) GOLD 5520+ (*2)	3
Server Memory		1024GB DDR5 (32GB *32)	
Server Network Type		Mellanox CX7 Single-port NDR IB 400Gb (*4)	
Data drives		DapuStor R6100 15.36TB NVMe SSD (*16)	

1.3. Switch

Mellanox(NVIDIA) Quantum-2 QM9700 switch, all clients and servers are connected to it.

1.4. Power & Rack Requirements

For detailed system power requirements, see the table below:

System component	Power supply unit	Nameplate Power(W)	Rated Power(W)	Design Power(W)	Rack Unit
Storage Node 1	Power supply 1	1300		1300	2
	Power supply 2	1300			
Storage Node 2	Power supply 1	1300		1300	2
	Power supply 2	1300			
Storage Node 3	Power supply 1	1300		1300	2
	Power supply 2	1300			
Totals	/	7800		3900	6

2. Software

2.1. Client

The detailed configuration of the client-side software is shown in the table below:

Software Type	Software Configuration/Version
Client OS	Ubuntu 24.04.4 LTS
Client Kernel	6.8.0-100-generic
UBIXFS	2.1.1
Mellanox OFED	MLNX_OFED_LINUX-24.10-4.1.4.0

2.2. Server

The detailed configuration of the server-side software is shown in the table below:

Software Type	Software Configuration/Version
Server OS	UbiPower OS
Server Kernel	2.1.1
Storage	UbiPower 18000
Mellanox OFED	MLNX_OFED_LINUX-24.10-4.1.4.0

2.3. Switch

NA

3. Settings

3.1. Client

- 1. Install standard UBIXFS POSIX client on the host with default settings.
- 2. The file system is mounted on the client using UBIXFS.
- 3. Limit the client's memory to 256GB via GRUB configuration.

3.2. Server

For detailed configuration information about storage servers, see the table below:

Storage server Configuration	Description
Vendor Name	UBIX TECHNOLOGY CO., LTD
Product Model	UbiPower 18000
Version	2.1.1
Redundancy Policy	EC 4+2:1

3.3. Switch

NA

4. Configuration

4.1. Storage config

`--rdma=true`

4.2. Client config

`--rdma=true --direct_io=true`